

FROM SETUP TO THE FINAL HEAT

Meet day. One heat at a time.

A practical guide to running Quantum Swimming. Prepare the system, follow the heat routine, and hand off checked results.

01 / BEFORE SWIMMERS RACE

Prepare & test.

Work with your timing lead. A saved setup still needs to match today's pool.

Manual pp. 5–7, 10–20

Pool first. Schedule second. Set the pool length, lanes and connected devices before importing or creating races.

01 Connect the equipment

Have the timing lead verify the start system, harnesses, touchpads, backup buttons, laptop, timer, scoreboard and results connection. Connect the **USB protection dongle**; races cannot start without it.

▼ Power and installation notes

Use approved power equipment and protect electronics from splashes. This guide assumes the hardware is installed. The manual refers physical installation to **Quantum Concept 3480.508.02**.

02 Open Quantum Swimming

On Home, check the timer indicator: **green means connected**; red means disconnected. In **Configuration** → **Timer**, choose **Auto** for the USB serial port. Allow about 10 seconds, or more than a minute for the first connection.

03 Synchronize the timer

For one timer, select PC time or enter a manual time, then use **Do synchro**.

▼ Using primary and secondary timers?

Choose **On Start Pulse**. Connect and power the start device. Enter the same desired time on both timers, press **Do synchro** on both, then send a start pulse. After the ready light returns, send a test pulse to verify synchronization.

04 Create or open the session

From Home, choose **Race**. Use a clear meet name without special characters, such as clubMeets1. The manual recommends **one Quantum meet per session**. Double-click an existing meet to open it.

05 Match the pool configuration

Check pool length, finish side, lane count and order, the module assigned to each lane, and the devices connected at each end. **A green module number means it is communicating.** Configure only the equipment actually fitted.

▼ Different pool length or yards?

The manual shows 25 m and 50 m options. Have the timing lead verify the supported configuration for your pool length and units before entering races.

06 Test every pad and backup button

Open **Pool test**. Ask a helper to touch each pad and press each backup button separately at every equipped end. Confirm the **correct lane and device counter increases**. Test platforms if fitted. Clear totals before the final check after warm-up.

Resolve missing or wrong-lane signals before racing.

07 Check timing settings and outputs

Have the timing lead check arming delays, start arming delay, warnings, printers and I/Os. Delays must suit your swimmers and pool. Enable the checkbox for every I/O you use. Check battery levels, printer paper and the team's backup timing arrangements.

Load the session.

Choose your workflow, then compare the loaded races with the meet program.

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WITH MEET MANAGEMENT SOFTWARE

Import the schedule

1. Enable the DH connection.

In I/Os, select the results team's interface. Agree the folder or connection settings and enable its checkbox.

2. Choose Import schedule.

Click the DH button above the workspace. Check the events, heats and lane assignments. Use the DH dashboard to import start lists if needed.

3. Verify the session.

Check event numbers, distances, strokes, rounds, heat counts, swimmers and scratches against the meet program.

WITHOUT A DATA CONNECTION

Enter races manually

1. Add the events.

Use Schedule's add-event control. Check distance, stroke, category and round; use edit-event to correct them.

2. Add each heat.

Select the event, then use add-heat once per heat needed.

3. Assign swimmers to lanes.

Double-click a lane to open Add Swimmer. Find or create the swimmer, select the lane, then choose Add to Startlist.

▼ HY-TEK connection settings

Select **DH_HYTEK**. Set the shared-folder path and HYTEK Data Set number agreed with the results operator. The manual specifies:

- `quantum.sch` — schedule
- `startlist.slx` — start lists

- *.qag — exported results

Confirm the installed HY-TEK version supports this exchange, then test import and export. DH_LSTFILES and DH_SPLASH use different settings; these filenames do not apply to every interface.

▼ Empty lanes and relay teams

With **Auto unused lanes** enabled, a blank name in a populated start list can mark the lane unused. Confirm the lane's actual status with the results team. When a heat has no swimmers assigned at all, all lanes open as in use.

For relay races, **Display relays** lets you enter the team and its swimmers in swimming order.

Do a practice run

Use a separate practice meet and a test results folder or data set agreed with the results operator. Open a heat, confirm the ready light, send a start and simulate touches in the correct order after the arming delays. Review times, practice using a backup, mark the test official and confirm delivery.

Return to the real session. Recheck the meet, pool, outputs, schedule and first heat before telling the starter you are ready.

03 / REPEAT FOR EVERY HEAT

The heat routine.

Follow the eight instructions in order. These guide controls do not operate Quantum.

Manual pp. 23–31

INSTRUCTION 1 OF 8 · BEFORE THE START

Open the correct heat.

On Schedule, select the event and double-click the heat. Read the **event number, heat number and distance** aloud with your partner. Check swimmers, active lanes and the expected finish end.

Check before moving on

The heat on screen matches the swimmers about to race.

INSTRUCTION 2 OF 8 · BEFORE THE START

Confirm the system is ready.

Start armed should be green and checked. The connected start input should be green. Confirm the starter has the ready light; a connected scoreboard should show **0.00**.

If an authorized official asks you to hold, click Start armed to disarm it. Re-arm only when cleared.

Check before moving on

Correct heat. Start armed. Starter ready light on.

INSTRUCTION 3 OF 8 · DURING THE RACE

Check that the clock starts.

After the start signal, confirm the race clock is running. Click the timing-window **focus button** so keyboard shortcuts reach the timing window.

Look for a yellow window frame

Yellow means the timing window has focus. Red means it does not.

INSTRUCTION 4 OF 8 · DURING THE RACE

Watch the lanes and touches.

Follow the touches, distance completed, direction and finish indicators. Compare the screen with the swimmers in the pool. Ask your timing lead about a missing touch or unexpected finish.

Keep the harness connected

Its disconnect control blocks incoming timing pulses.

INSTRUCTION 5 OF 8 · AFTER THE FINISH

Review every lane.

In Result view, confirm the **event, heat and final distance**. This matters if the timing and result windows have been unlinked. Check finish times, missing results and differences from backup times.

Confirm statuses with the official

DNS = did not start · DNF = did not finish · DSQ = disqualified.

INSTRUCTION 6 OF 8 · AFTER THE FINISH

Resolve questions before approval.

Check questionable pad times against recorded backups and the officials' decision. Record each correction and its reason.

Quantum's backup-difference indicators include **200 ms** comparisons. They are review cues, not automatic instructions to replace a result.

Check before moving on

Every missing time or disputed result has been reviewed.

INSTRUCTION 7 OF 8 · AFTER THE FINISH

Mark the heat official.

When review is complete, click the result's **official control** in Quantum. Depending on enabled settings, this may print, change the scoreboard or export results.

Confirm the intended output. Use the DH export action if your configured workflow requires it.

Confirm the handoff

The results operator has received the checked result.

INSTRUCTION 8 OF 8 · AFTER THE FINISH

Advance to the next heat.

In Quantum, use **Next heat** or **Ctrl** + **N**. After an event's last heat, it advances to the next event. Resolve unfinished or unofficial warnings before continuing.

Start the routine again

Read the new event number, heat number and distance aloud.

Need to recover a missed start or missing touch? [Open troubleshooting](#).

SAY IT TOGETHER

Before: "Event 12, heat 3, 100 freestyle. Lanes checked. Ready."

After: "All lanes reviewed. Results confirmed. Official. Next heat."

Find the problem.

Choose what you see. Involve the timing lead in changes to race data.

Manual pp. 5–9, 23–31

The timer indicator is red

1. Check timer power and the USB connection.
 2. In **Configuration** → **Timer**, check Auto or the correct serial port. Allow about 10 seconds; the first connection can take more than a minute.
 3. If it remains disconnected, ask the timing lead to troubleshoot. Restore readiness before the next heat.
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A genuine start happened while Quantum was unarmed

1. Open the correct heat, if needed.
2. Inspect the time beneath **Attrib last start**. If it is the genuine race start, use the button to apply it.
3. The adjacent arrow lists other recorded unarmed starts. Have the timing lead identify the correct signal. A later test pulse must not become the race start.

Do not reset a valid race to fix a missed start. Recover the recorded start.

The officials recall a start after the clock begins

On the officials' instruction, use **Arm start** in Quantum to reset timing for another start. Confirm the correct heat and ready light again. This is different from recovering a genuine start that happened while unarmed.

A pad touch or finish is missing

1. Identify the lane and expected end. Check module communication and whether the pad was armed.
 2. The backup-time button, when shown, or **B** in the lane menu takes the recorded backup time.
 3. Review the evidence with the officials and follow the meet's approved timing procedure before marking the result official.
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The lap or distance progress is wrong

Click the lane number to open its menu. After checking the actual swim, use add-lap or remove-lap as needed. Recheck the distance and expected next touch.

The white control above the lane buttons can change **all swimmers**. Use it only if the same correction is required for every lane.

A lane needs to be armed or marked unused

Use its lane menu or pad-arm indicator. In the lane menu, arms the pad, disarms it and toggles used/unused on the manual's US keyboard layout. Verify the selected lane first.

A missing name can automatically mark a lane unused. Check the start list before changing it.

The scoreboard or results computer is not updating

1. Check the relevant I/O checkbox is enabled.
 2. Verify the selected output, port or shared-folder path with the results operator.
 3. Confirm which heat is displayed in Result view before sending. Use scoreboard refresh or DH export as appropriate, then verify the destination.
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The protection dongle is missing

Check that the USB protection key is connected; without it, races cannot start. Ask the timing lead for the licensing procedure in **manual section 4.3.1**.

Keyboard shortcuts do not respond

Click the timing-window focus button and confirm a **yellow frame**. Use visible controls if your keyboard mapping differs from the manual's US layout.

05 / KEEP BESIDE THE COMPUTER

Meet-day checklist.

Check off the preparation, repeat the heat checks, and finish the session.

0 of 16 checks complete

Before the first heat

- ☐ Correct meet open; timer connected and synchronized.
- ☐ Pool configured before loading races; every pad and button tested.
- ☐ Schedule, names, lanes and distances checked against the meet program.
- ☐ Starter, scoreboard, results transfer and backup timing tested.

Every heat

- ☐ Verify event, heat, distance, lanes and finish end.
- ☐ Confirm Start armed, connected start input and ready light.
- ☐ Check the running clock; give the timing window focus.
- ☐ Watch touches and distance; investigate anything unexpected.
- ☐ Review the correct heat and final-distance results.
- ☐ Resolve corrections, mark official and confirm result delivery.
- ☐ Select Next heat and repeat the pre-start checks.

At the end of the session

- ☐ Resolve unfinished or unofficial heats with officials.
- ☐ Confirm the results team has all final results and corrections.
- ☐ Close the race window; exit through Home → Quit.
- ☐ Copy session data and exchanged result files to the team's backup location; verify the copy.
- ☐ Hand off equipment issues and follow the product care instructions.

▼ Where to find data and how to finish equipment care

The manual's usual data location is %stDrcApp%\Quantum\Swimming\Data\[meet name]. Have the timing lead locate the actual folder on the computer.

Clean touchpads with fresh water and a cloth, let them dry naturally and store vertically. Follow each product's instructions for the remaining equipment. Backing up the session and checking the copy are recommended team practices.

06 / KNOW WHAT YOU ARE LOOKING AT

Quick reference.

Words used in Quantum

Event	A race category, such as 100 freestyle.
Heat	One group swimming an event.
Start list	Swimmers assigned to each lane.
DH	Data handling: the connection to meet-management software.
SCB	The scoreboard connection.
Official	Quantum's reviewed-result status; it may also trigger printing or export.

Heat status letters

I	Idle	W	Waiting for start
S	Started	F	Finished, unofficial
O	Official	U	Unfinished

Useful shortcuts

Ctrl + **N** Next heat

In a lane menu: **B** Backup time · **Q** Arm pad · **Y** Disarm pad

These apply inside Quantum using the manual's US keyboard layout. The timing window must have focus.

▼ Suggested team roles

Timing operator: runs Quantum. **Results operator:** checks the meet-management system and receives results. **Deck helper:** tests pads and buttons. **Starter and referee:** handle race decisions.

Small teams may combine roles. Agree responsibilities and the backup timing process before the meet.

Source and scope. Based on Swiss Timing's *QUANTUM-SWIMMING User's Manual*, 3480.509.02, version 1.4, October 2019. References use printed manual page numbers; add 6 to find the PDF page. Labels and icons may differ in your installed version.

This is a team learning aid. Team roles, practice drills and sign-off checks are suggested operating practices. The guide's checkboxes and walkthrough do not connect to or control your timing equipment.

[Open the Swiss Timing manual ↗ \(Swiss Timing manual 3480.509.02\)](#)
